ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 6 - Due date 02/27/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A06_Sp26.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "ggplot2", "forecast", "tseries" and "sarima". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here
#install.packages("sarima")
libraries <- c("tidyverse", "forecast","tseries","sarima", "here")
lapply(libraries, require, character.only=TRUE)
```

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Loading required package: forecast
##
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
##
## Loading required package: tseries
```

```
##
## Loading required package: sarima
##
## Loading required package: stats4
##
##
## Attaching package: 'sarima'
##
##
## The following object is masked from 'package:stats':
##
##     spectrum
##
##
## Loading required package: here
##
## here() starts at C:/Users/jhsal/OneDrive - Duke University/797/TSA2026/TSA_Sp26
```

```
## [[1]]
## [1] TRUE
##
## [[2]]
## [1] TRUE
##
## [[3]]
## [1] TRUE
##
## [[4]]
## [1] TRUE
##
## [[5]]
## [1] TRUE
```

```r
here()
```

```
## [1] "C:/Users/jhsal/OneDrive - Duke University/797/TSA2026/TSA_Sp26"
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

  Answer: The ACF plot of an AR(2) series would demonstrate a slow, positive decay, while the PACF plot would show 2 significant lags. Seasonality would be apparent but it is not determined by the AR(2) distinction

- MA(1)

  Answer: the ACF plot of an MA(1) series would show one significant negative lag, and a long tail in the PACF. Seasonality would be apparent but it is not determined by the MA(1) distinction

Q2

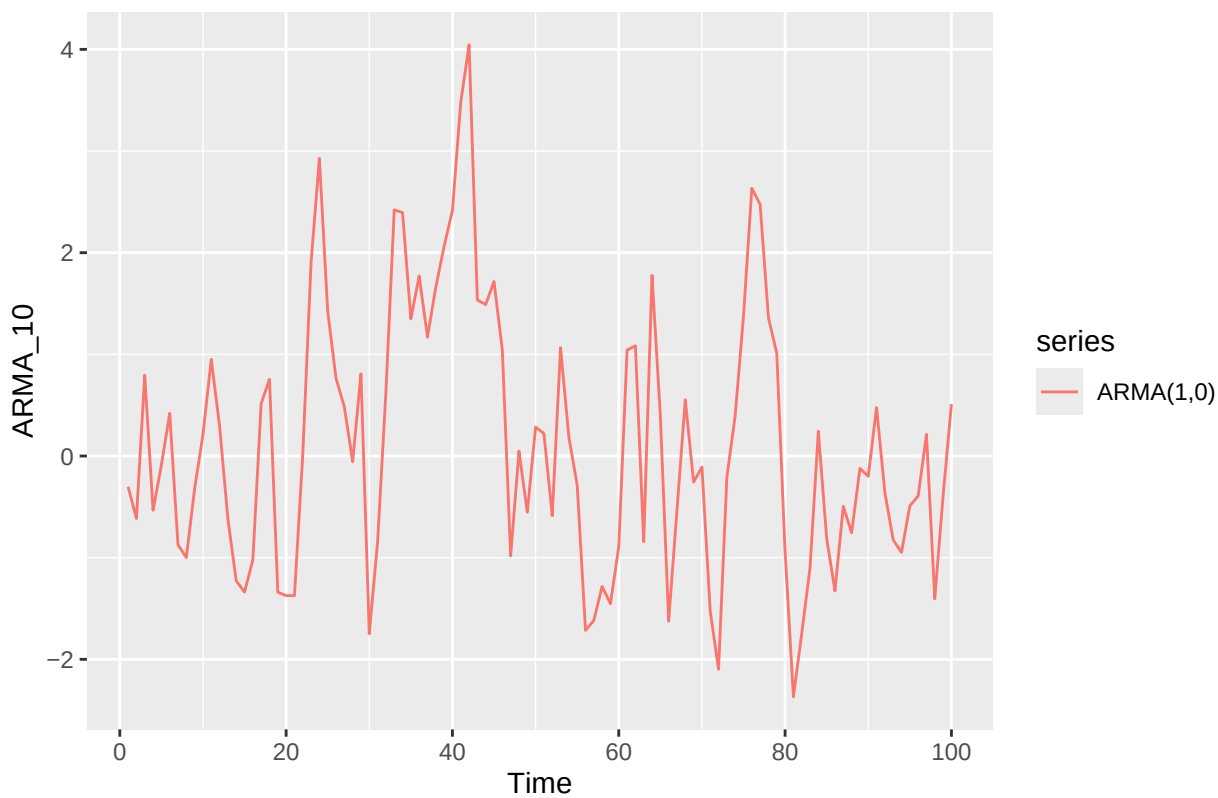
Recall that the non-seasonal ARIMA is described by three parameters ARIMA(p, d, q) where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the ARMA(p, q).

(a) Consider three models: ARMA(1,0), ARMA(0,1) and ARMA(1,1) with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arima.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

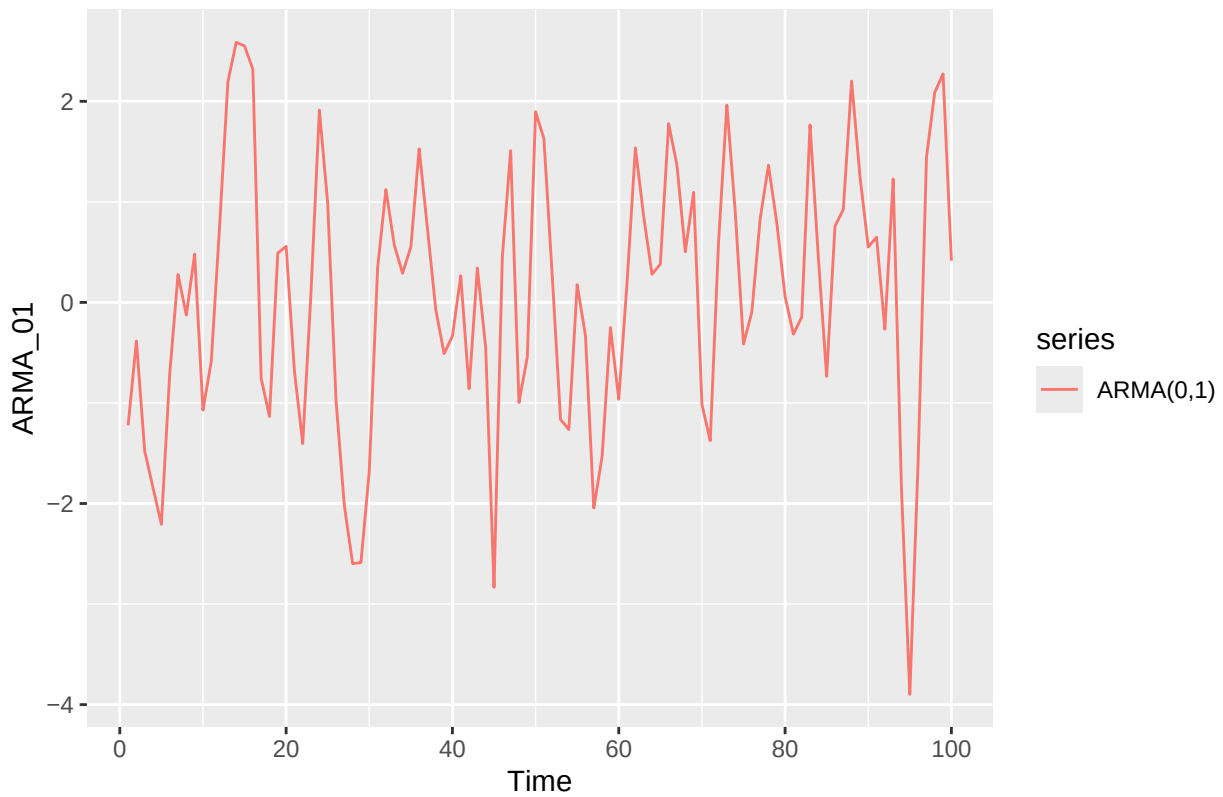
```
ARMA_10 <- arima.sim(n = 100, list(ar = (0.6), ma = 0))
ARMA_01 <- arima.sim(n = 100, list(ar = (0), ma = 0.9))
```

```
## Warning in min(Mod(polyroot(c(1, -model$ar)))): no non-missing arguments to
## min; returning Inf
```

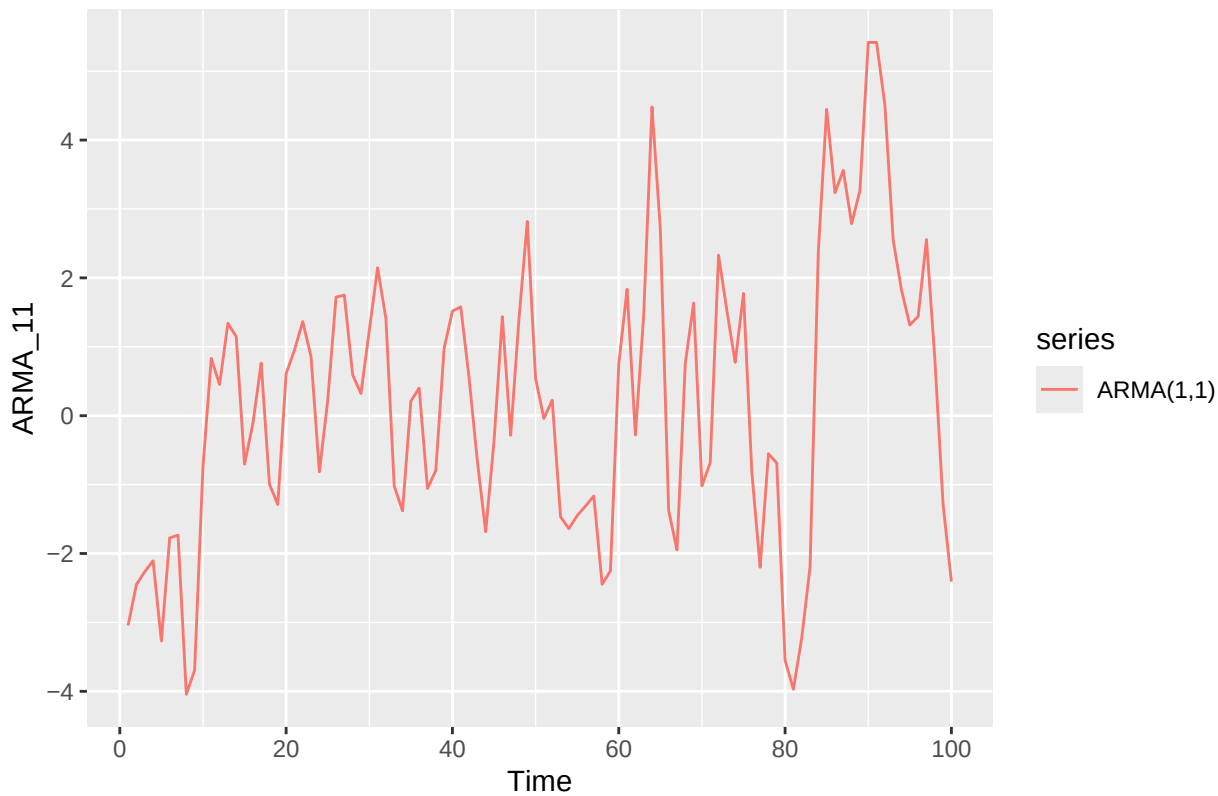
```
ARMA_11 <- arima.sim(n = 100, list(ar = (0.6), ma = 0.9))

autoplot(ARMA_10, series = "ARMA(1,0)")
```




```
autoplot(ARMA_01, series = "ARMA(0,1)")
```

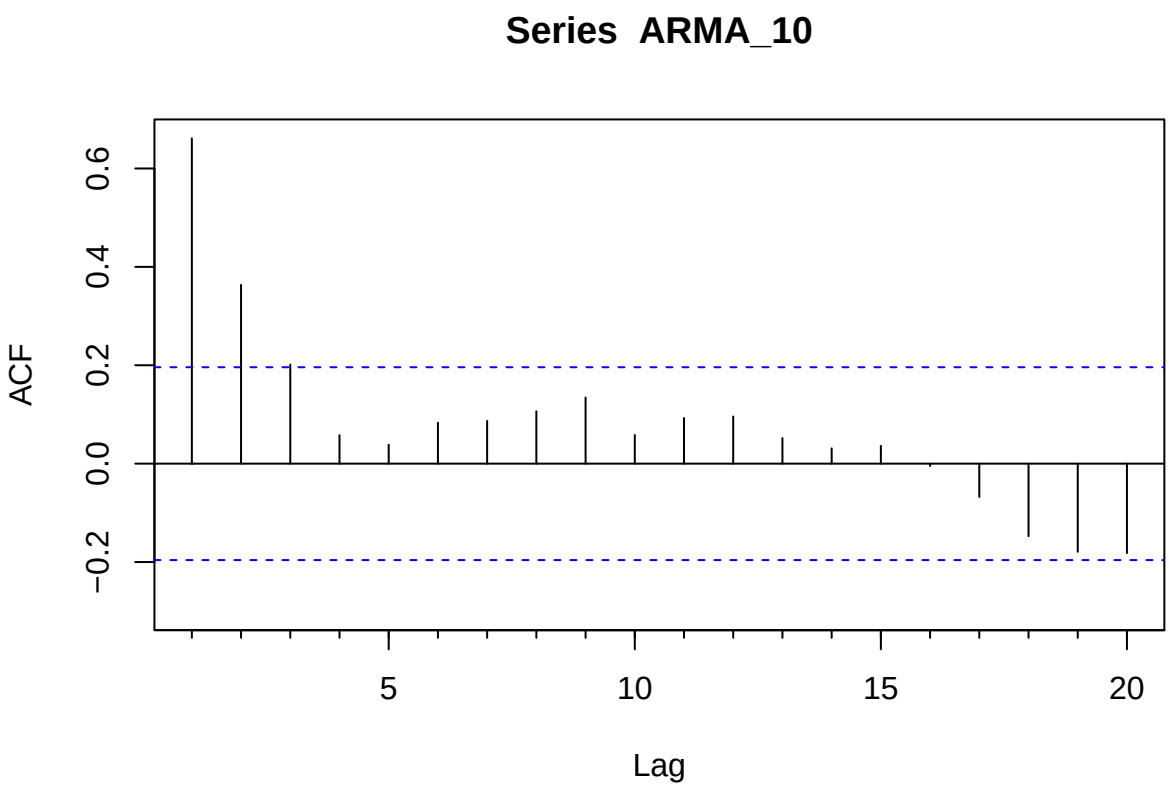


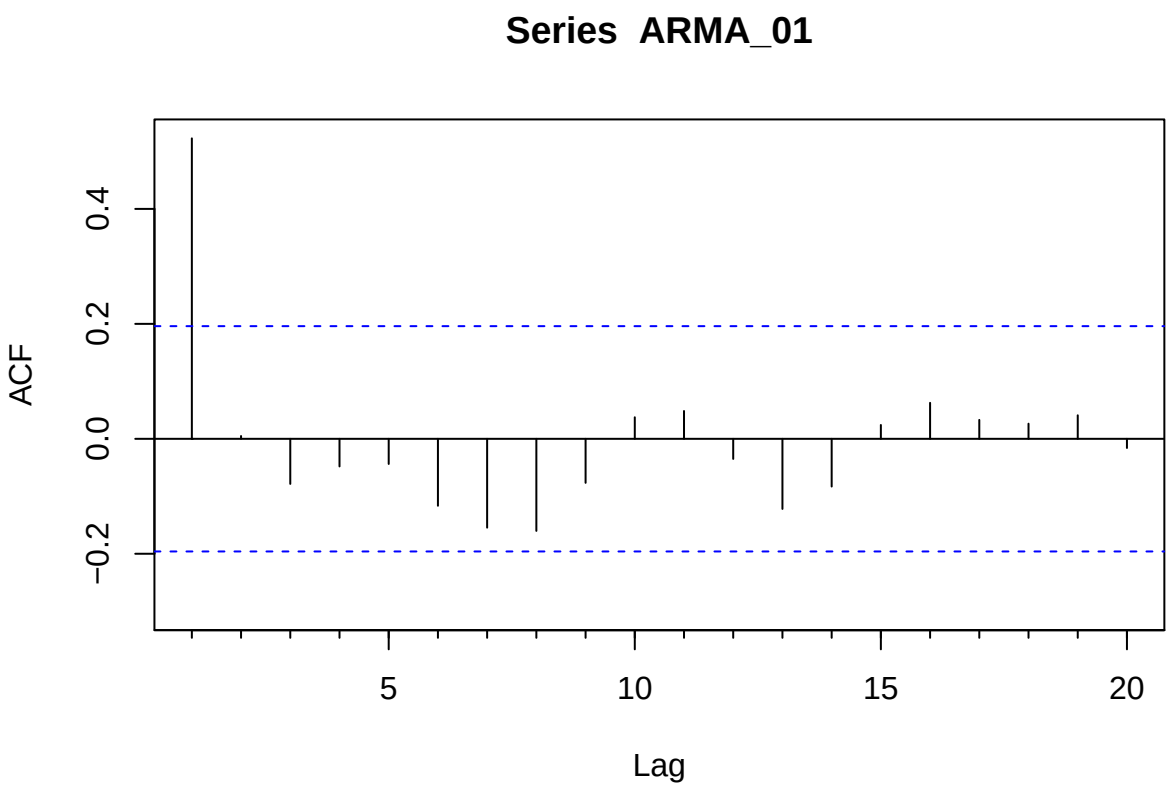

```r
autoplot(ARMA_11, series = "ARMA(1,1)")
```

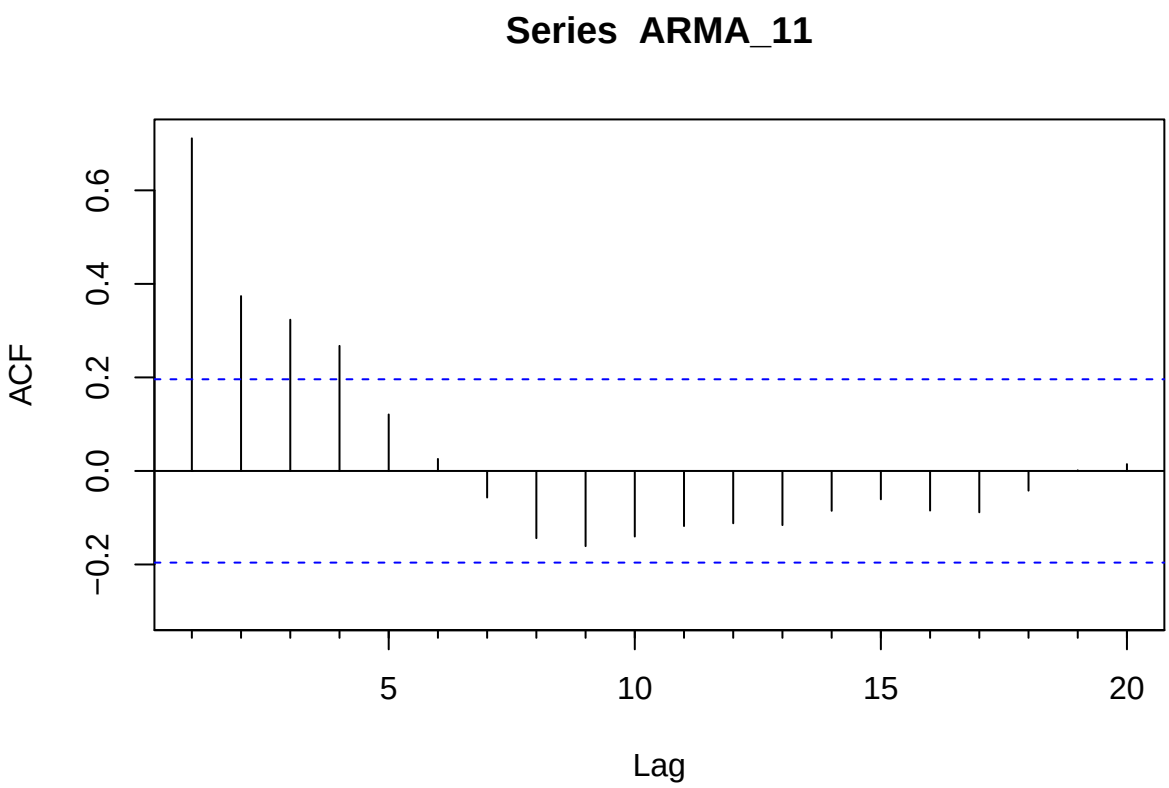


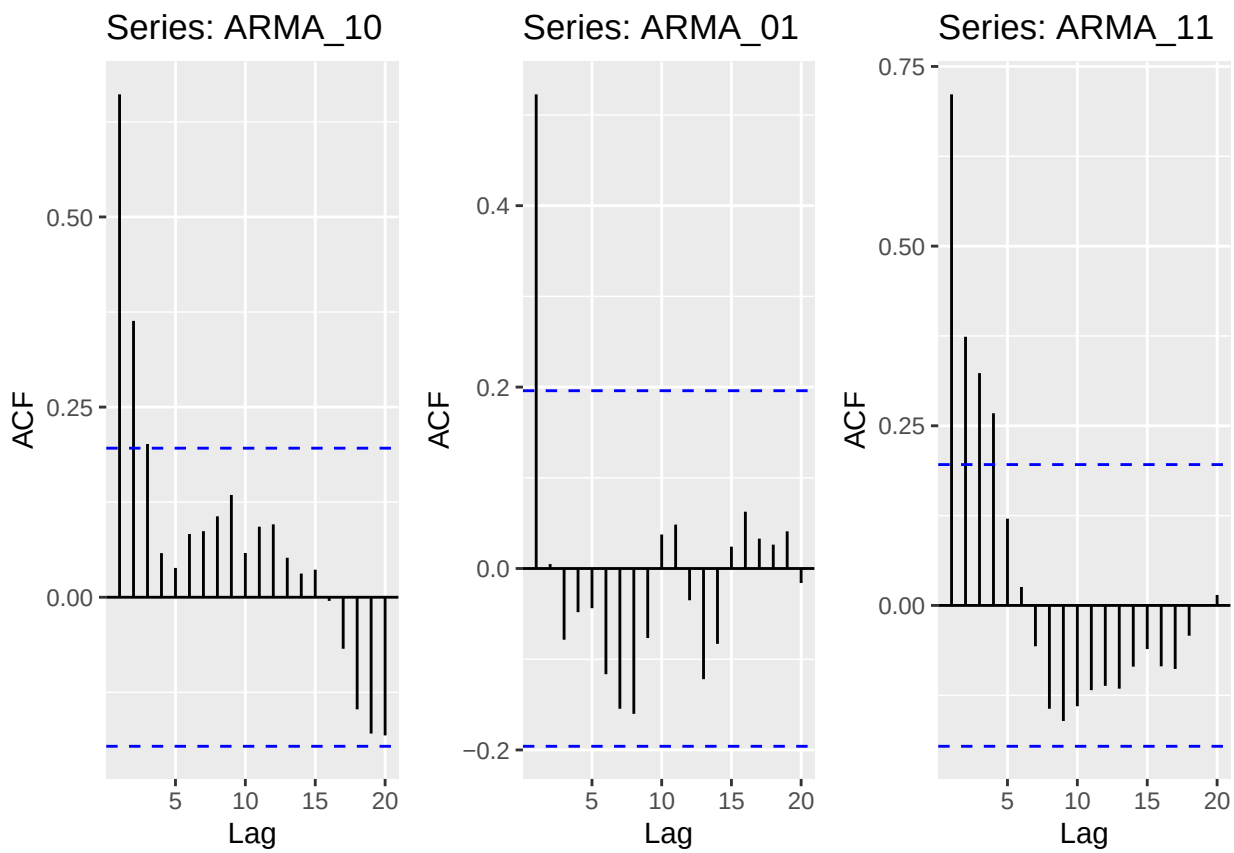
(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

```
cowplot::plot_grid(
  autoplot(Acf(ARMA_10)),
  autoplot(Acf(ARMA_01)),
  autoplot(Acf(ARMA_11)),
  ncol = 3
)
```


Series ARMA_10
ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag

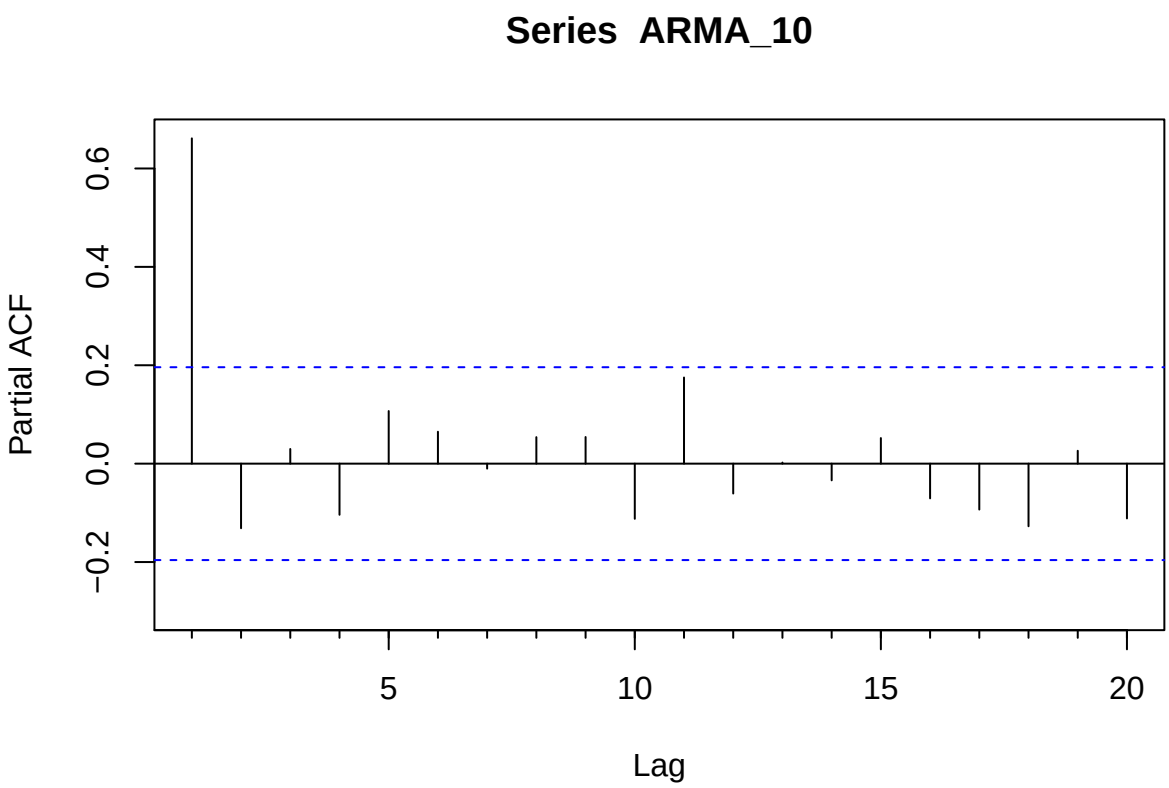

Series ARMA_01
ACF
0.4
0.2
0.0
−0.2
5
10
15
20
Lag

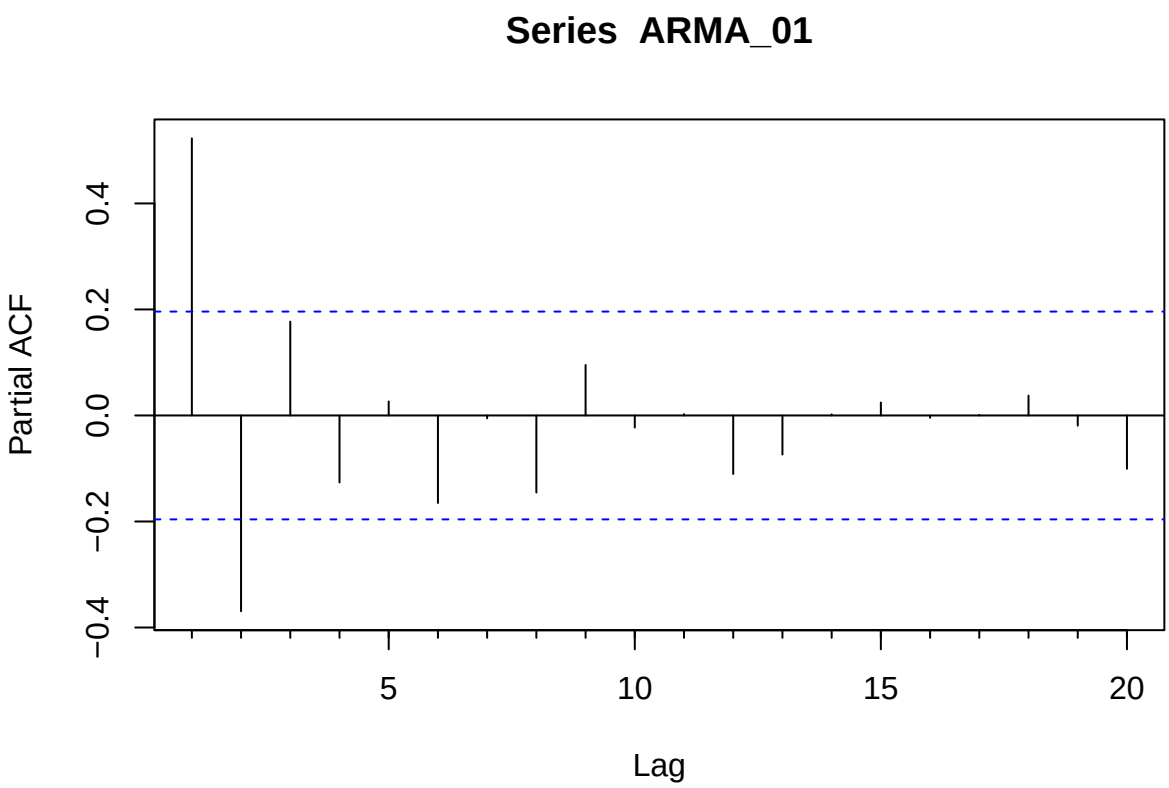

Series ARMA_11
ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag

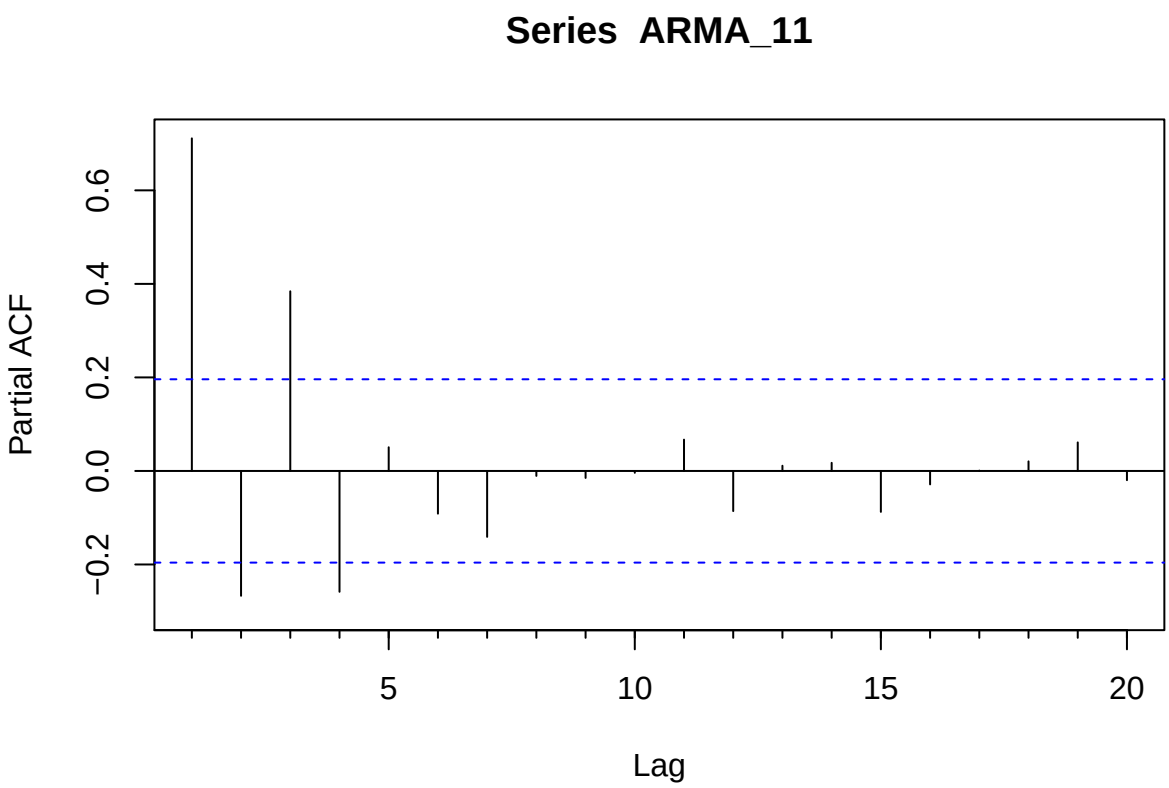


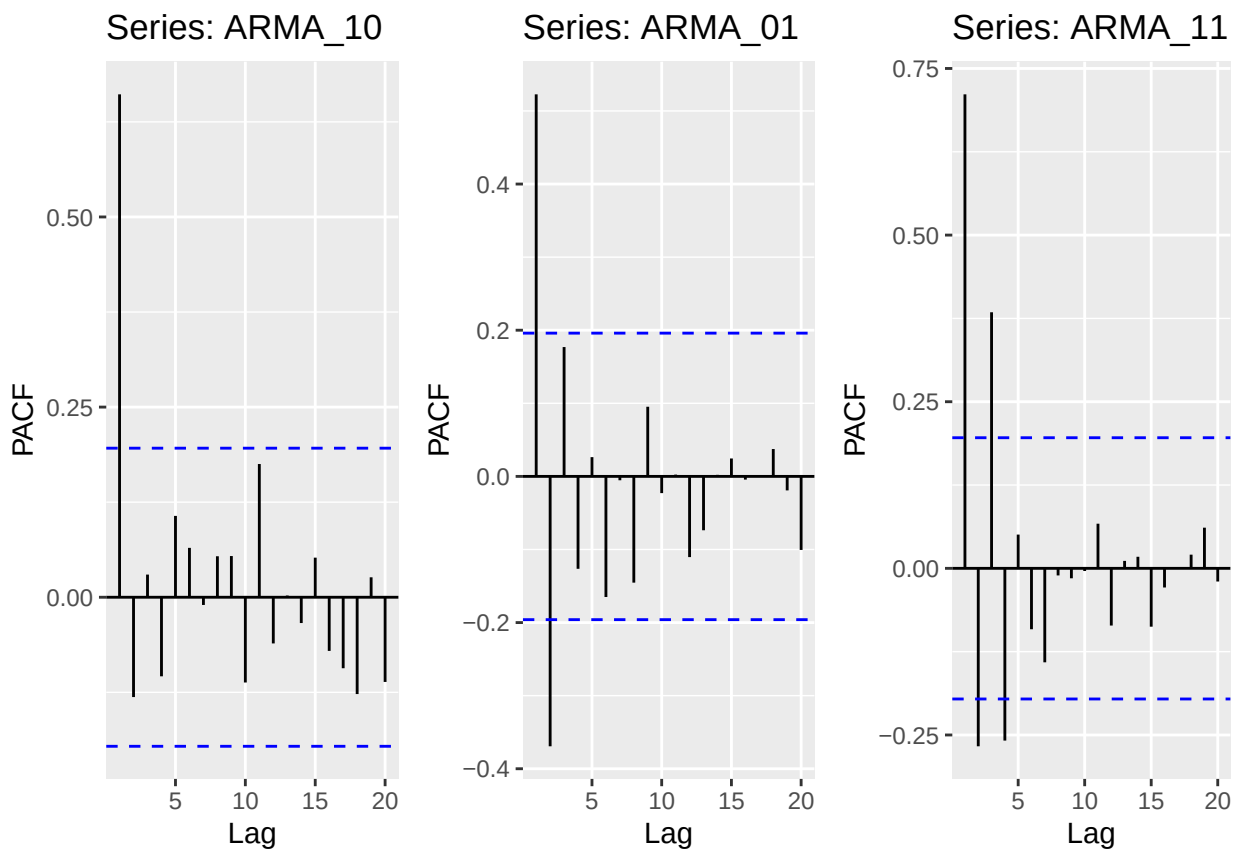
(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
cowplot::plot_grid(
  autoplot(Pacf(ARMA_10)),
  autoplot(Pacf(ARMA_01)),
  autoplot(Pacf(ARMA_11)),
  ncol = 3
)
```


Series ARMA_10
Partial ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag


Series ARMA_01
Partial ACF
0.4
0.2
0.0
−0.2
−0.4
5
10
15
20
Lag


Series ARMA_11
Partial ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag



(d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able identify them correctly? Explain your answer.

Answer: Yes, the ACF & PACF plots were each representative of the kinds of models we generated. In the case of ARMA(1,0), the ACF had a slow decay, and a single significant lag, indicating significant temporal dependence on $t-1$. ARMA(0,1) demonstrated only one significant lag in the ACF and a strong negative lag in the PACF, both of which are representative of the moving average function. The ARMA(1,1,) exhibits tendencies of both kinds of plots: a slow decay in the ACF and a strong negative lag in the PACF

(e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

Answer: in the ARMA(1,0) function, the .coefficient of 0.6 appears, because this is the measure of the exact correlation of the first lag to the current value. However, the ARMA(1,1) model does not match because of the distortions imposed by the moving average values.

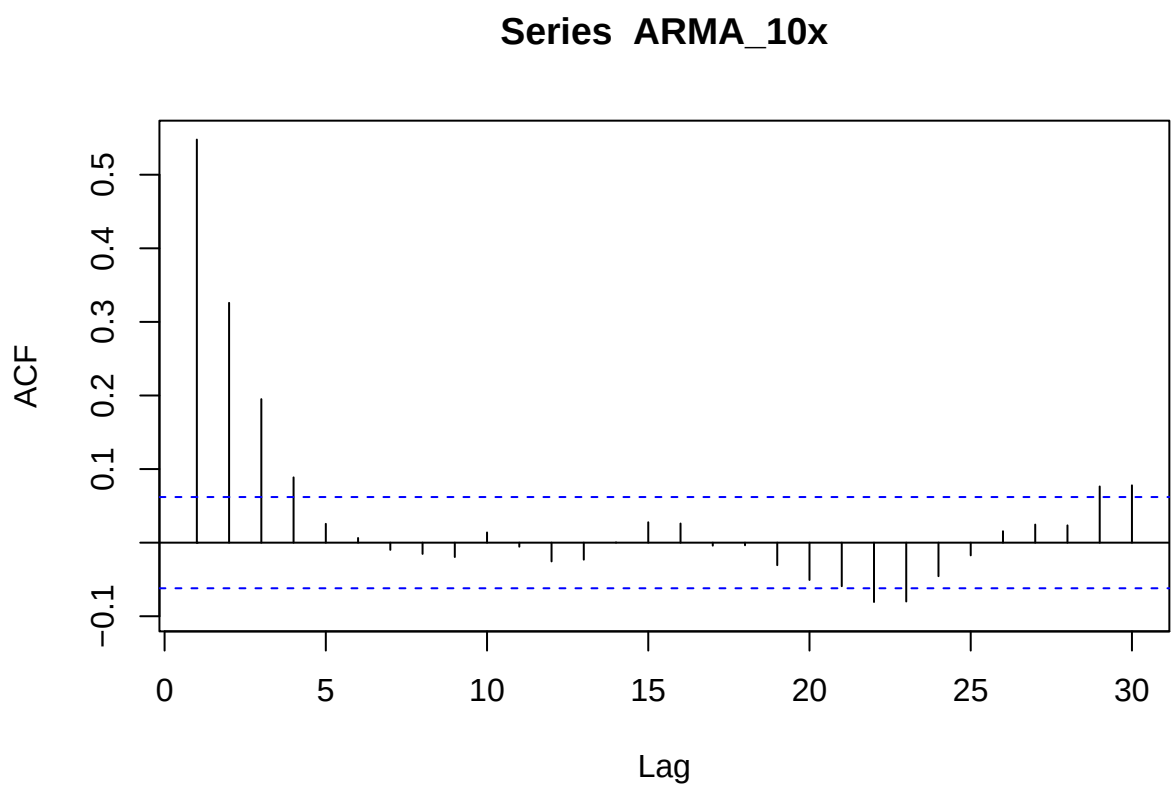
(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

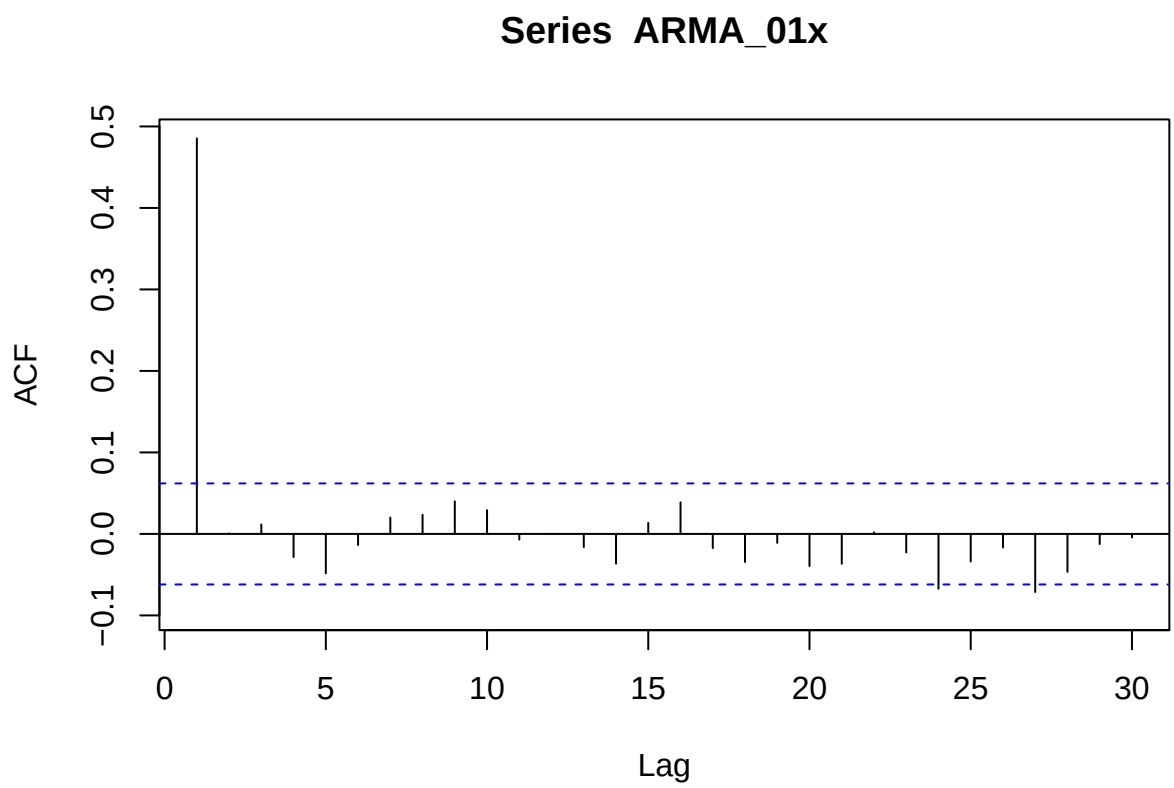
```
ARMA_10x <- arima.sim(n = 1000, list(ar = (0.6), ma = 0))
ARMA_01x <- arima.sim(n = 1000, list(ar = (0), ma = 0.9))
```

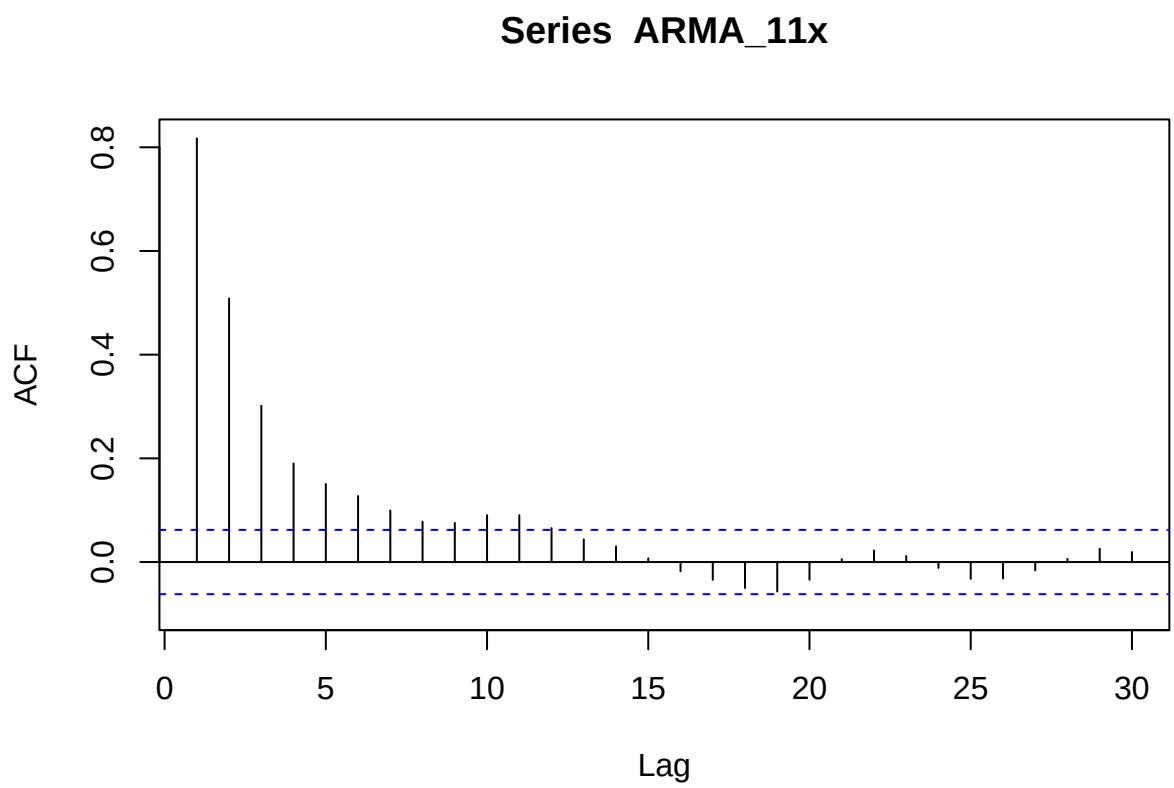
```
## Warning in min(Mod(polyroot(c(1, -model$ar)))): no non-missing arguments to
## min; returning Inf
```

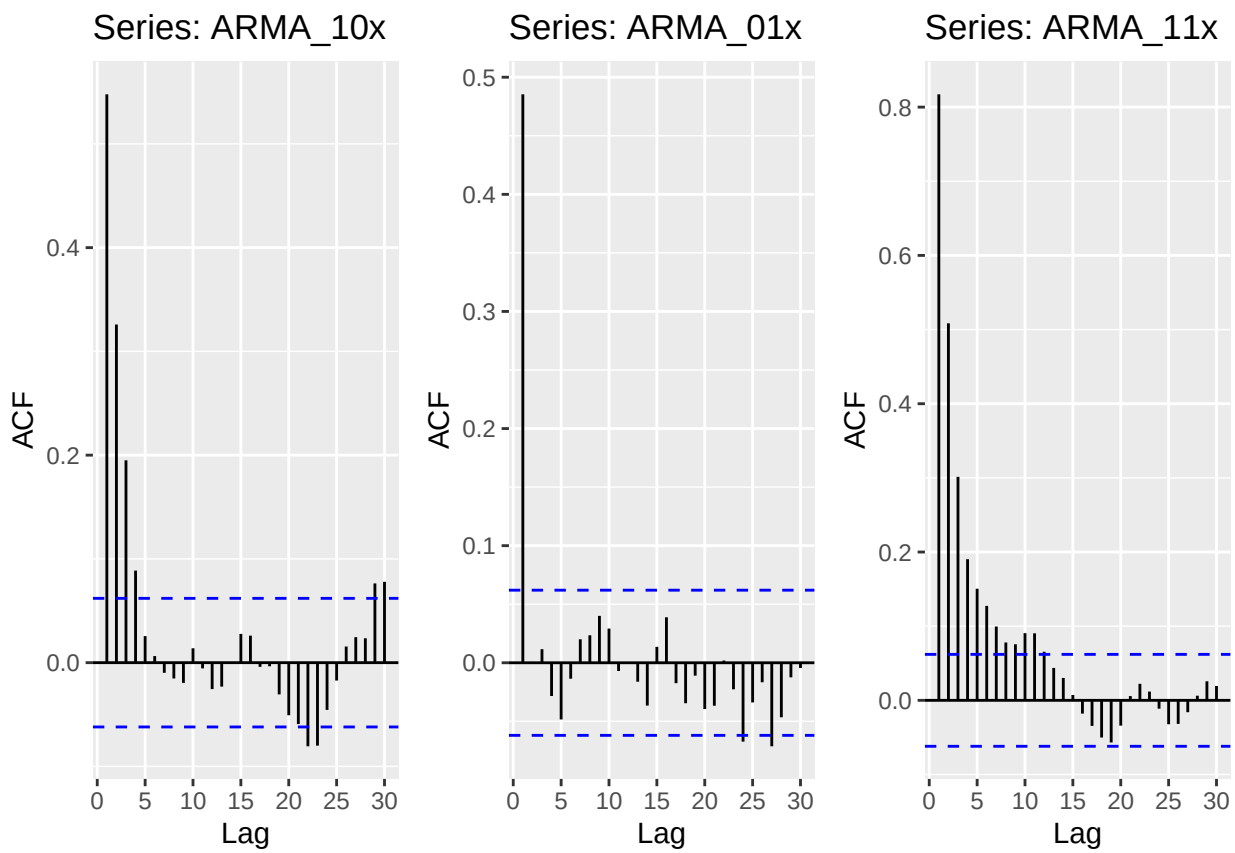
```
ARMA_11x <- arima.sim(n = 1000, list(ar = (0.6), ma = 0.9))
```

```r
cowplot::plot_grid(
  autoplot(Acf(ARMA_10x)),
  autoplot(Acf(ARMA_01x)),
  autoplot(Acf(ARMA_11x)),
  ncol = 3
)
```

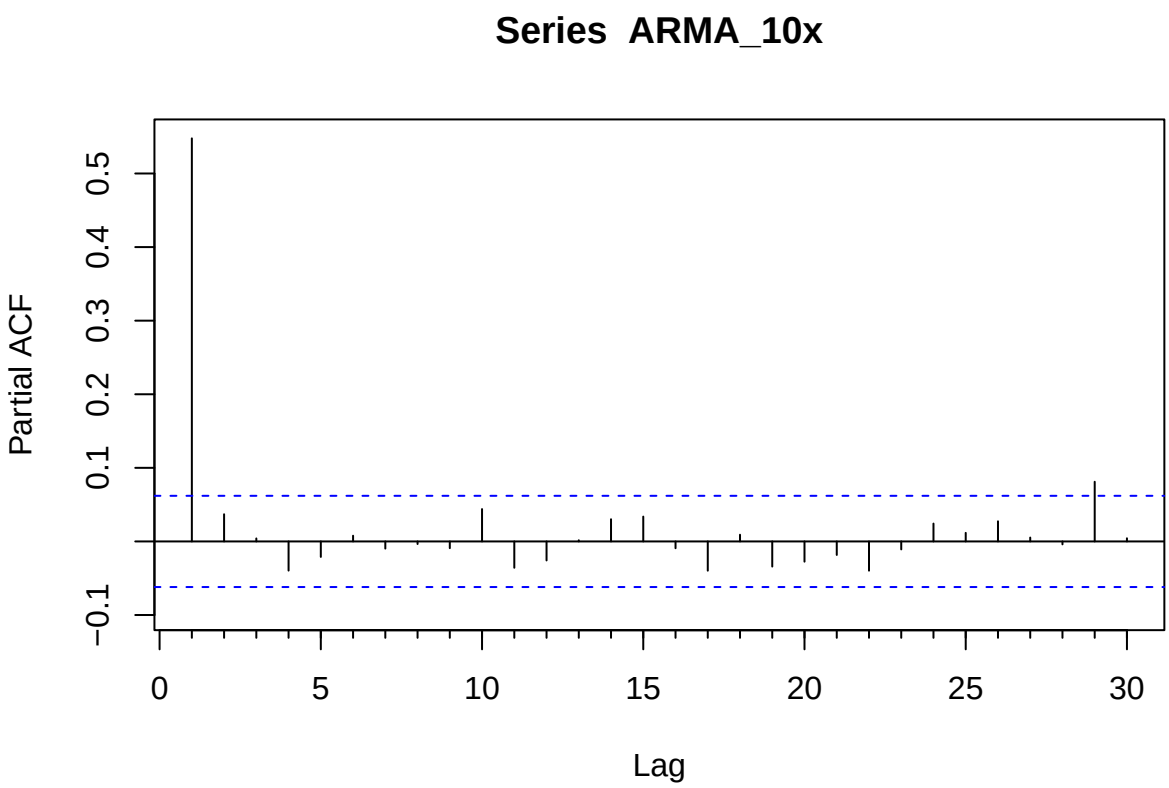


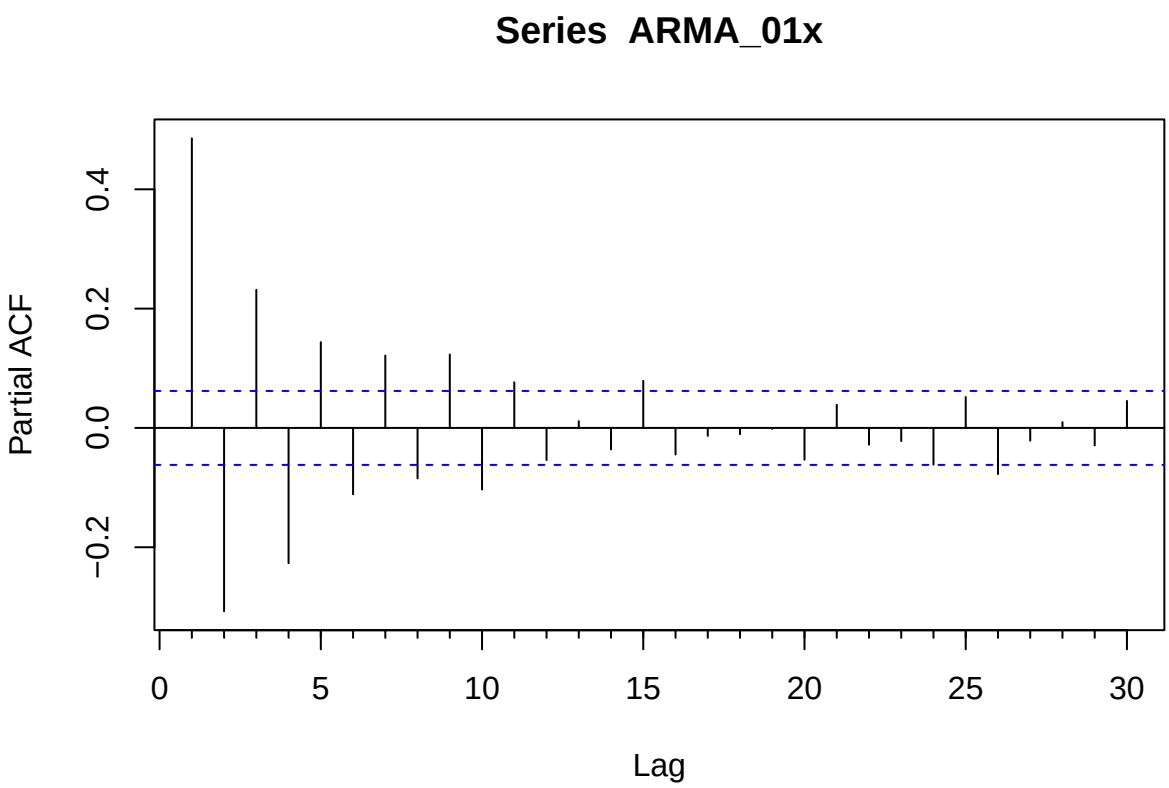

Series ARMA_01x
ACF
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0
5
10
15
20
25
30
Lag

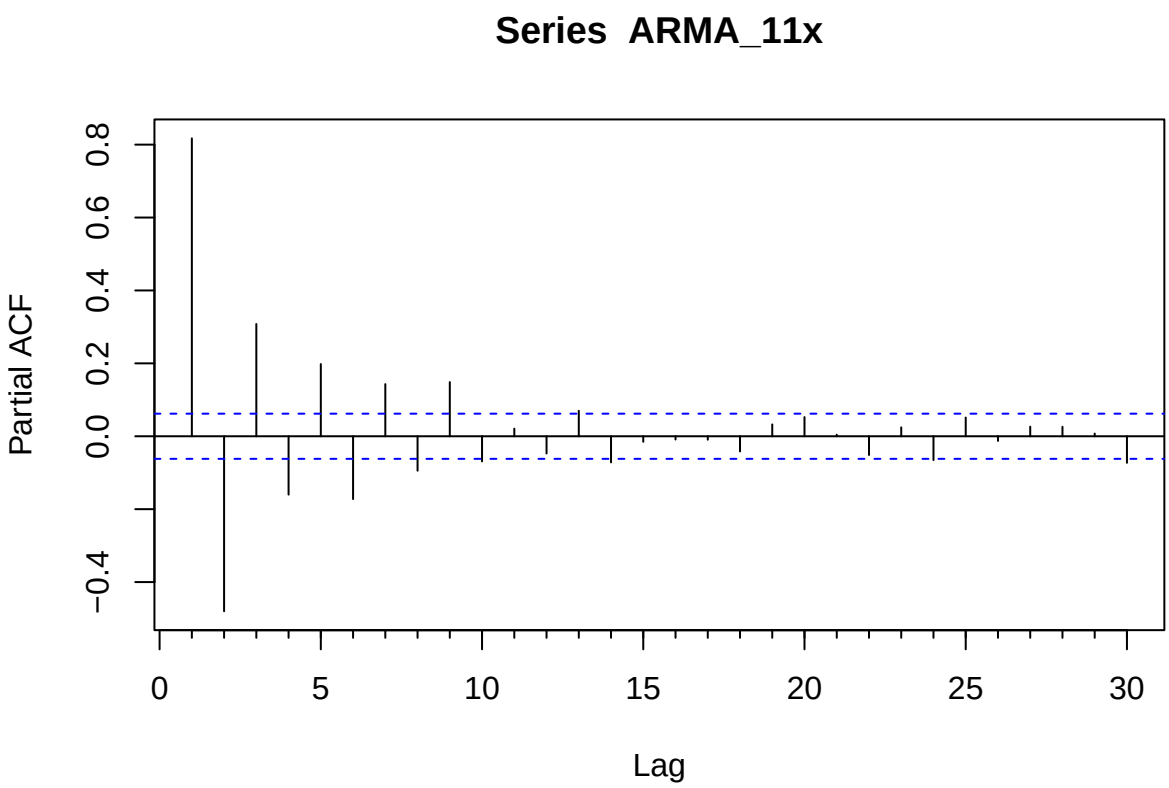

Series ARMA_11x
ACF
0.0
0.2
0.4
0.6
0.8
0
5
10
15
20
25
30
Lag

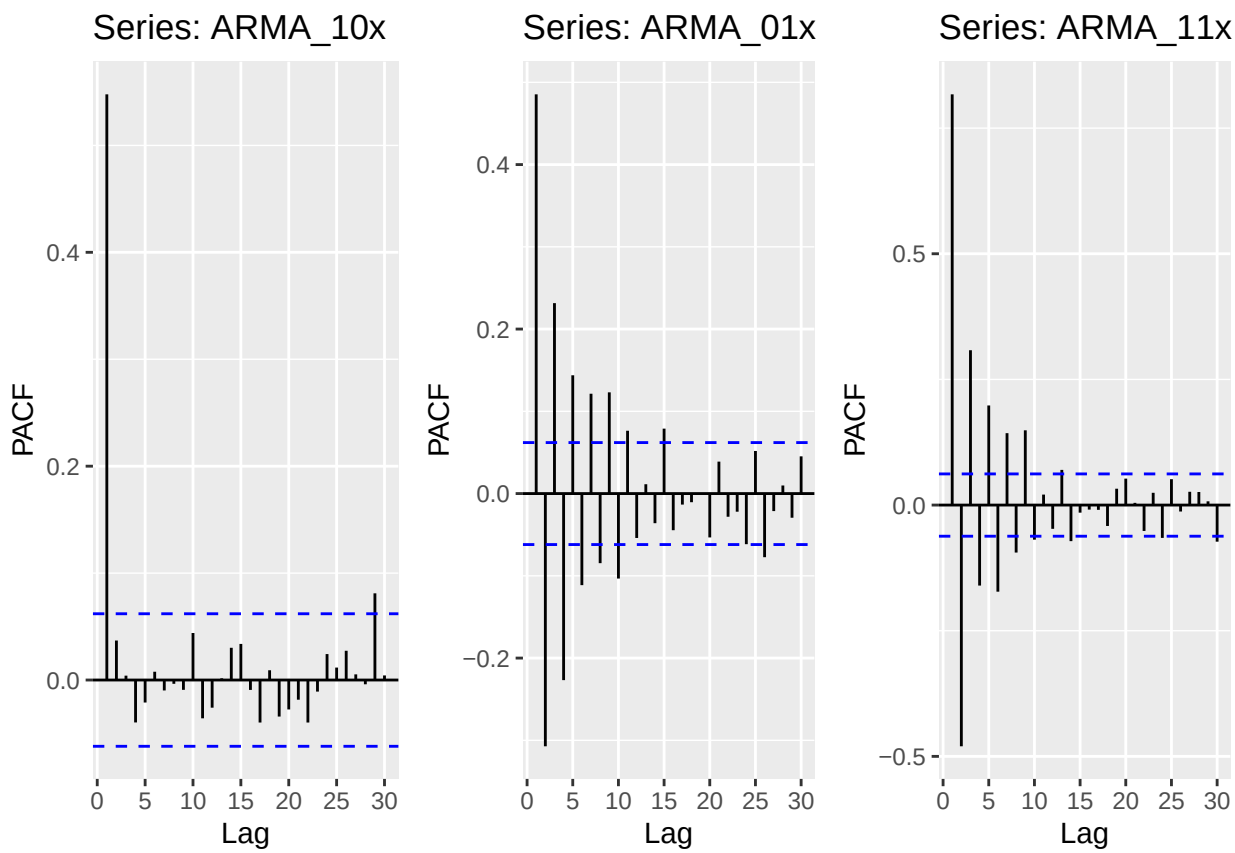



```
cowplot::plot_grid(
  autoplot(Pacf(ARMA_10x)),
  autoplot(Pacf(ARMA_01x)),
  autoplot(Pacf(ARMA_11x)),
  ncol = 3
)
```


Series ARMA_10x
Partial ACF
0.5
0.4
0.3
0.2
0.1
−0.1
0
5
10
15
20
25
30
Lag


Series ARMA_01x
Partial ACF
0.4
0.2
0.0
−0.2
0
5
10
15
20
25
30
Lag


Series ARMA_11x
Partial ACF
0.8
0.6
0.4
0.2
0.0
−0.4
0
5
10
15
20
25
30
Lag



Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

(a) Identify the model using the notation ARIMA$(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

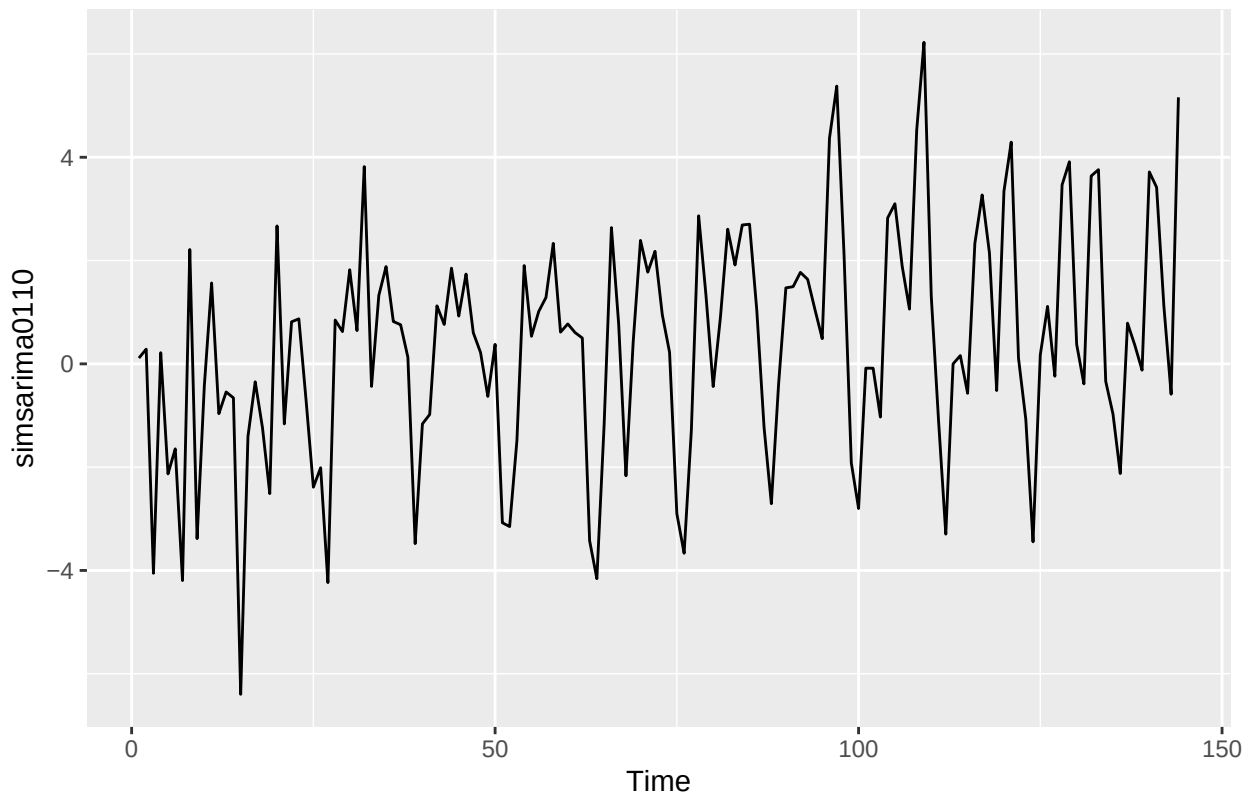
(b) Also from the equation what are the values of the parameters, i.e., model coefficients.

answer: ARIMA(1,0,1)(1,0,0). The model coefficients are $\phi_1 = 0.7$, $\phi_{12} = -0.25$, $\theta_1 = -0.1$

Q4

Simulate a seasonal ARIMA$(0, 1) \times (1, 0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

```
library(sarima)
simsarima0110 <- sim_sarima(n = 144, model = list(ma = 0.5, sar = 0.8, nseasons = 12))
simsarima0110 <- ts(simsarima0110)
autoplot(simsarima0110)
```

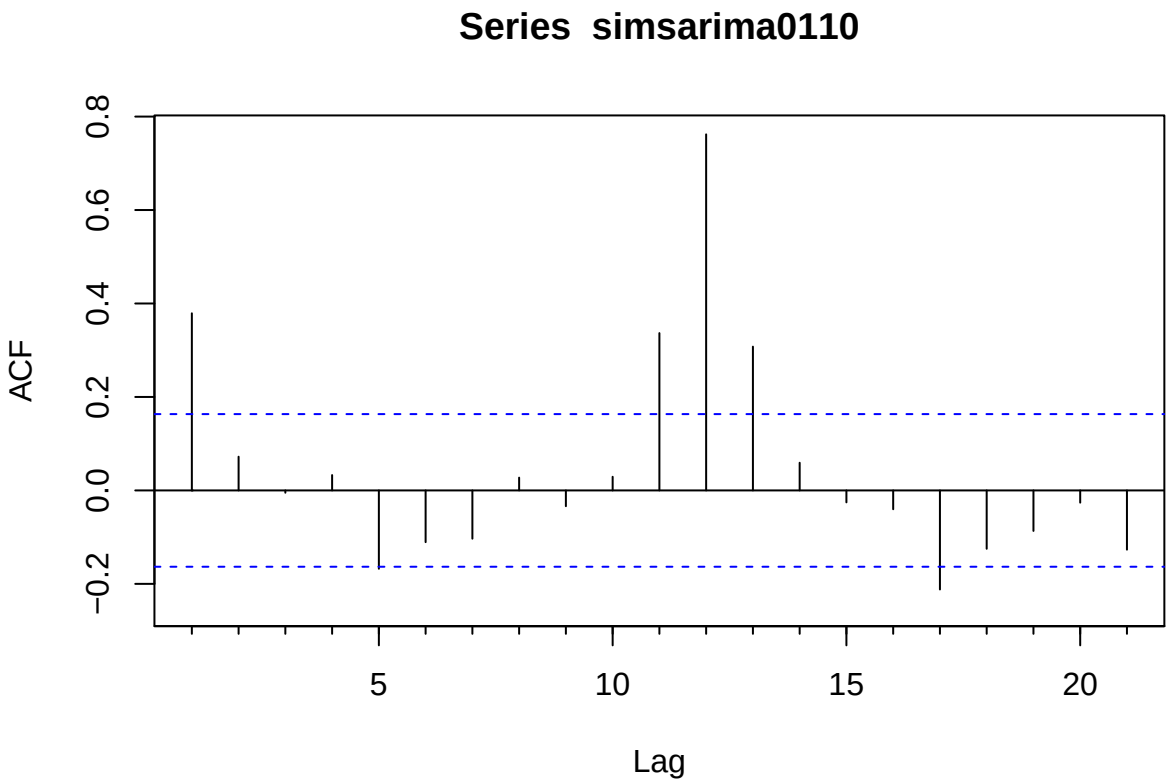


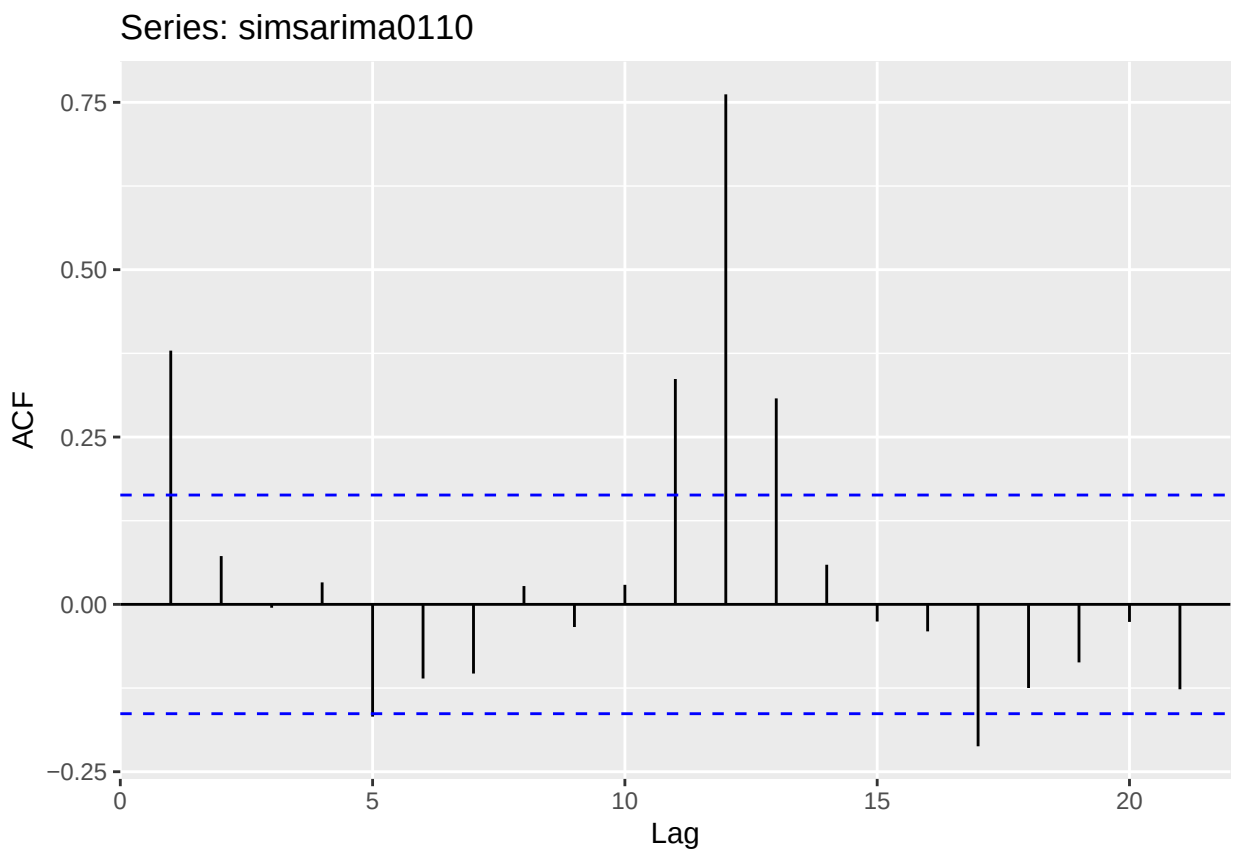
Answer: this plot does not have a strong, regular pattern of seasonality. The moving average component is quite strong compared to the seasonality, though we do see some patterns of peaks and patterns of troughs.

Q5

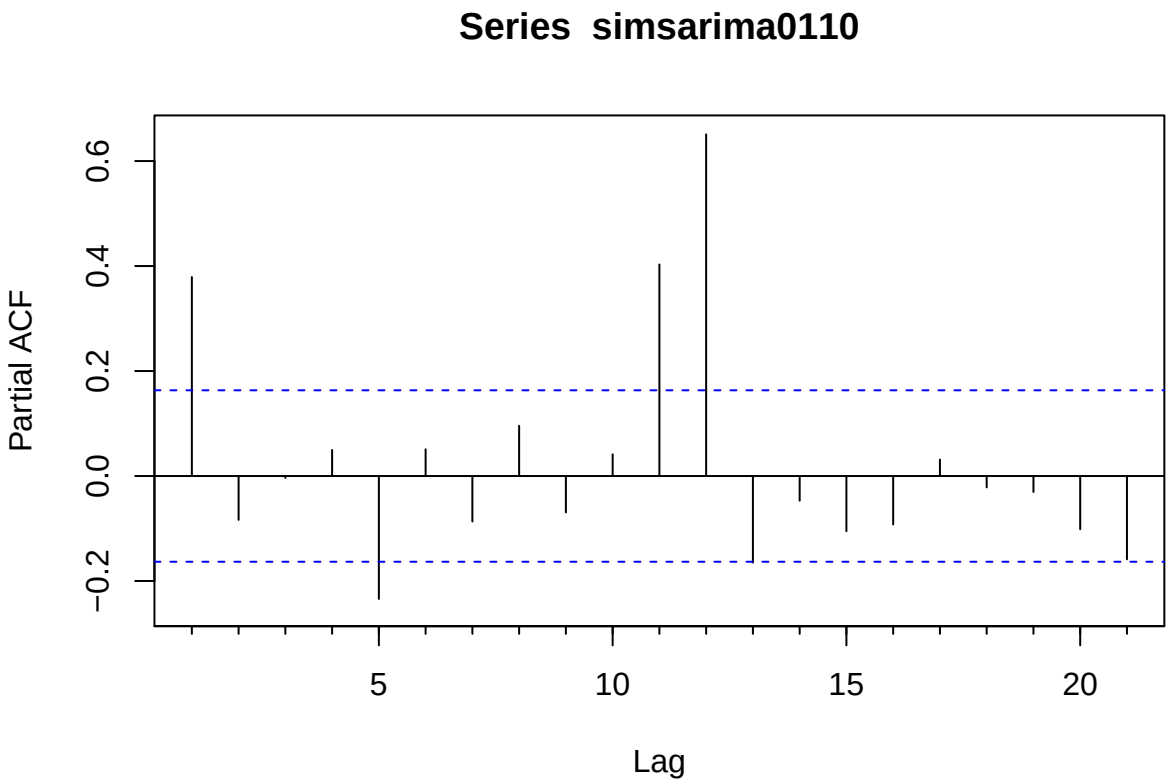
Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.

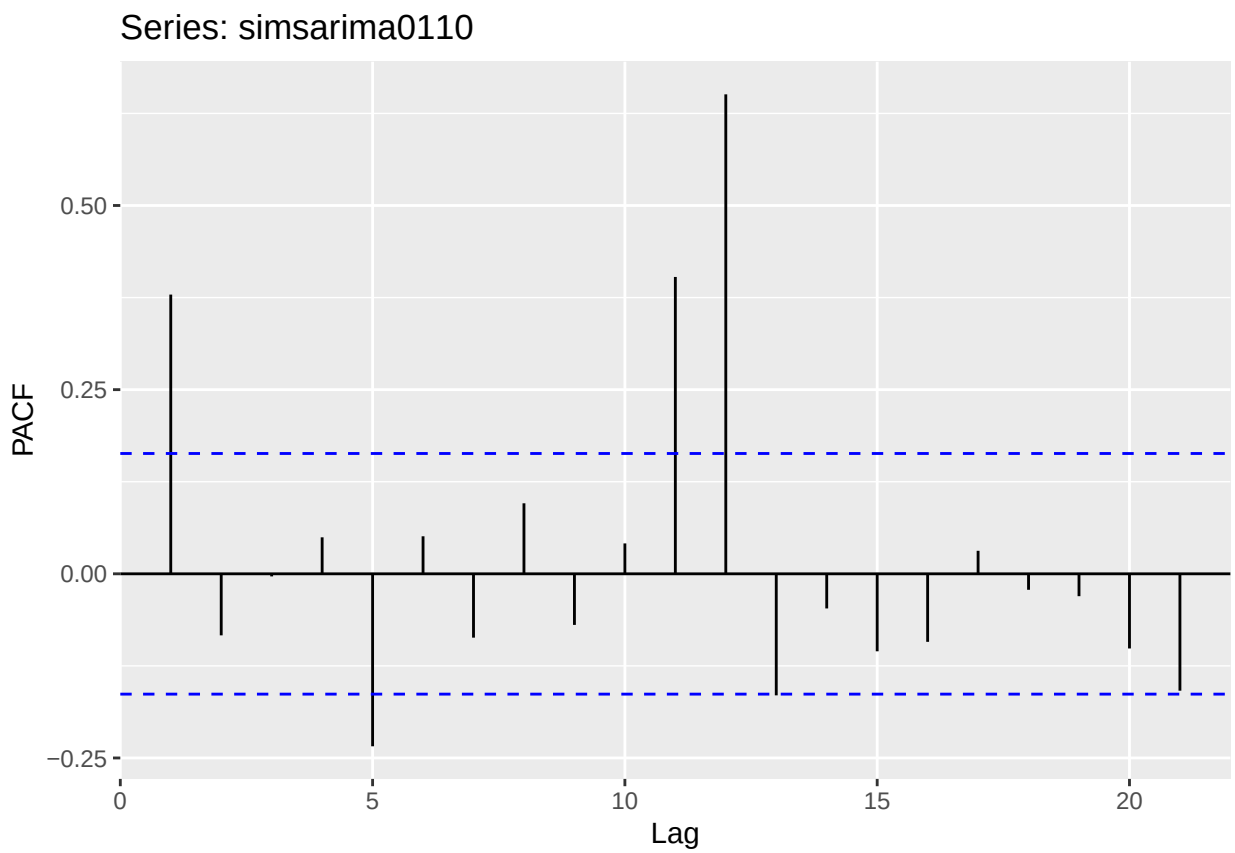
```
autoplot(Acf(simsarima0110))
```


Series simsarima0110
ACF
0.8
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag




```
autoplot(Pacf(simsarima0110))
```


Series simsarima0110
Partial ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
20
Lag



Answer: These graphs are very challenging to discern the exact structure of the series. Because there are interesting irregularities, such as spikes at surprising junctures, we can assume there is something going on, but I personally would have a lot of trouble being confident in identifying either the non-seasonal or the seasonal components of the series.